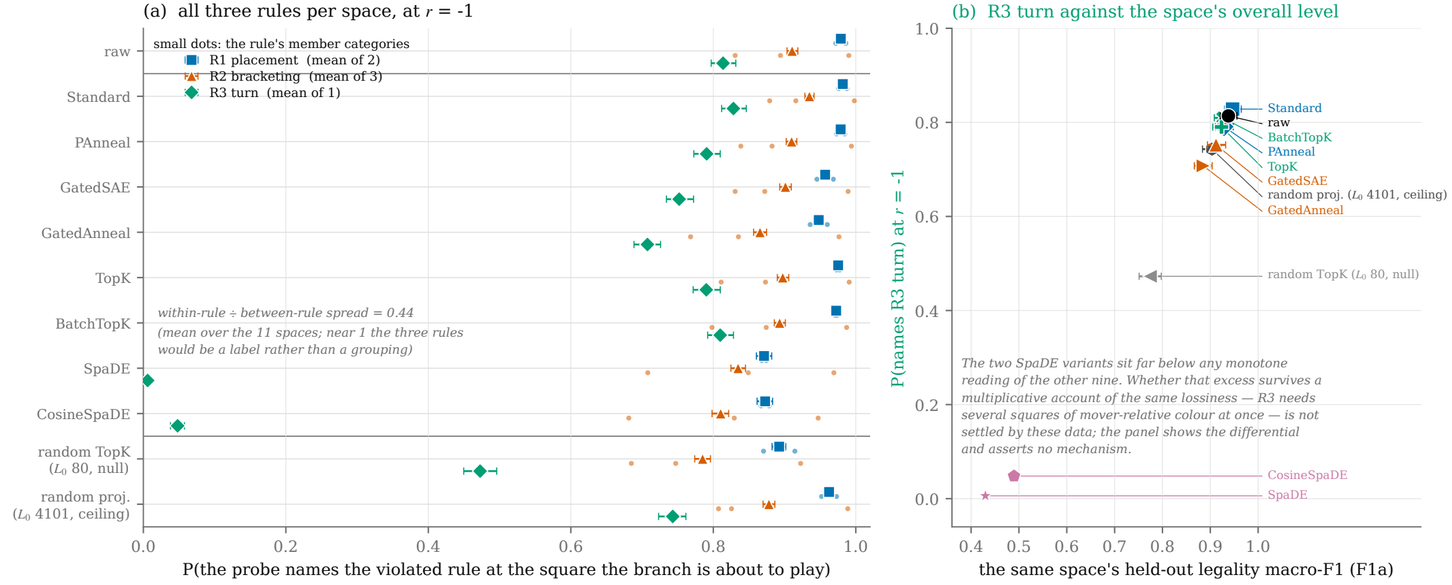
Read at r = -1, the board every branch of a game shares bitwise, one ply before the illegal move: this is a property of the representation space, not an effect of the violation. The mover-relative frame alternates each ply, so R3 (turn) is near zero at even offsets by construction and -1 is the phase in which the branch's move is about to be played. bars: 95% bootstrap CI over source games (n = 1997-2000). Rows and markers are F1's and F2's; a one-member rule is labelled as such so it is not read as an average.



R1 placement — a move puts one disc of the mover's color on an empty square.
R2 bracketing — the disc must close a gapless line of opponent discs with one of the mover's own.
R3 turn — colors alternate; legality is judged for the mover, not the opponent.